

# RAMYA S

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## PROFESSIONAL SUMMARY

B.Tech graduate in Artificial Intelligence and Data Science with hands-on experience building machine learning, Generative AI, NLP, and computer vision applications using Python, PyTorch, Scikit-learn, YOLO11, LangChain, Gemini, and Qdrant. Experienced in RAG pipelines, model evaluation, semantic similarity, and deploying AI applications with Streamlit and APIs. Seeking entry-level AI/ML Engineer, AI Engineer, or Generative AI Engineer opportunities.

## TECHNICAL SKILLS

**Programming Languages:** Python, SQL, JavaScript

**Machine Learning:** Scikit-learn, Logistic Regression, Random Forest, Model Evaluation, Cross-Validation, EDA, Data Preprocessing

**Deep Learning & Computer Vision:** PyTorch, YOLO11, OpenCV, Pillow, Object Detection

**Generative AI & NLP:** Large Language Models (LLM), LangChain, Gemini, Qdrant, RAG, Prompt Engineering, Semantic Similarity, Sentence Transformers

**Data & Applications:** Pandas, NumPy, Matplotlib, Seaborn, Streamlit, Flask, REST APIs

**Tools:** Git, GitHub, VS Code, Postman, Vercel

## PROJECTS

**Industrial Vision AI Inspector** | Python, PyTorch, YOLO11, OpenCV, Streamlit

- Built a computer vision system to detect workers and PPE (helmets, safety vests) and generate worker-level safety compliance alerts using bounding-box spatial association.
- Trained a YOLO11n model (15 epochs, image size 640) on the Ultralytics Construction-PPE dataset, achieving Precision 0.695, Recall 0.462, and mAP@50 0.494; deployed live on Streamlit Cloud.

**TN Government Schemes RAG Assistant** | Python, LangChain, Gemini, Qdrant, Streamlit

- Built a bilingual (English/Tamil) Retrieval-Augmented Generation app answering questions on Tamil Nadu agricultural schemes with source and page references.
- Implemented Gemini embeddings, Qdrant vector search, and refusal logic; achieved a 100% source retrieval rate on 15 test questions, deployed live on Streamlit Cloud.

**AI Intelligence Pipeline** | Python, Async Programming, APIs, LLMs

- Built an asynchronous data pipeline collecting and enriching startup, product, and research-paper data (1,000 records each) with LLM extraction via Gemini → Groq → DeepSeek fallback.
- Implemented deterministic entity resolution with fuzzy matching, LLM fallback, and retry/backoff handling; designed the pipeline architecture to scale to 500K+ records.

## EXPERIENCE

**Artificial Intelligence Intern** — Alfido Tech | Aug 2026 – Present

- Worked on practical AI/ML assignments involving Python and machine learning concepts; developed and evaluated AI/ML solutions through structured, task-based projects.

**Machine learning Intern** — Garuda Aerospace | Feb 2026

- Worked on YOLOv9 and computer vision concepts for AI-based applications; assisted with data preprocessing and Python-based tasks.

**Data Science Intern** — Internz Learn | Apr 2024 – May 2024

- Streamlined project management processes, reducing data project delays by 20% and improving operational efficiency by 15% across teams.

**Additional Experience:** RISE Internship Program (Python, Flask/Django, REST APIs); Junior Java Developer — TATA STRIVE (Java, Spring Boot); Cybersecurity Analyst — TCS Forage (Virtual Internship)

## ACHIEVEMENTS & ACTIVITIES

- ZENTHRA AI · ICET 2026** — Presented technical work at an AI-focused technical event.
- StartupX Hackathon 2026 · Gamnexus** — Participated in Round 1: Startup Idea & PPT Submission.

## EDUCATION

**B.Tech, Artificial Intelligence and Data Science** — Agni College of Technology | 2022 – 2026 | CGPA: 8.3/10

## CERTIFICATIONS

- AWS – Solutions Architecture Job Simulation
- AI Tools Workshop
- Certificate of Presentation – ZENTHRA AI (ICET 2026)
- Certificate of Participation – ICICSDF'26